

Lecture Notes: Graph Theory—Eulerization

Math F113X Math and Society

Given a graph, when can you find:

(a) An Euler circuit?

Every vertex is even degree

(b) An Euler path? (that is not a circuit)

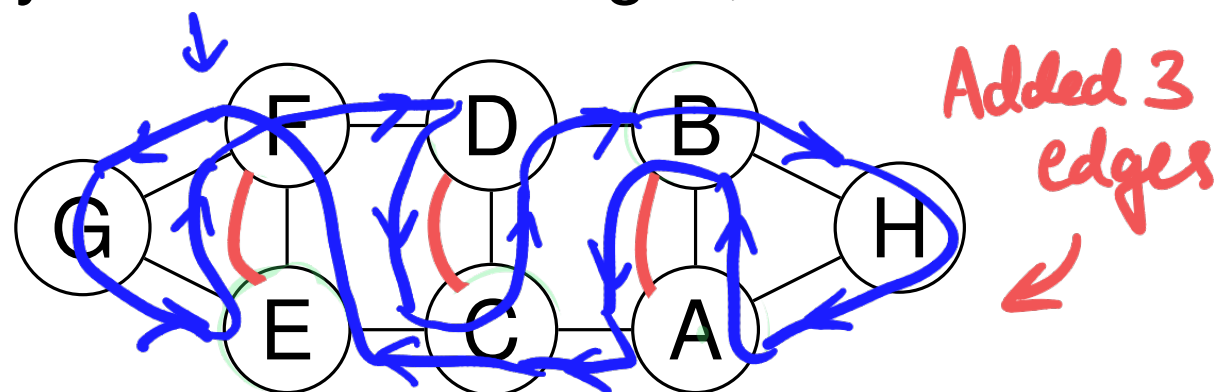
There are two vertices of odd degree
(start & end)

(c) Neither?

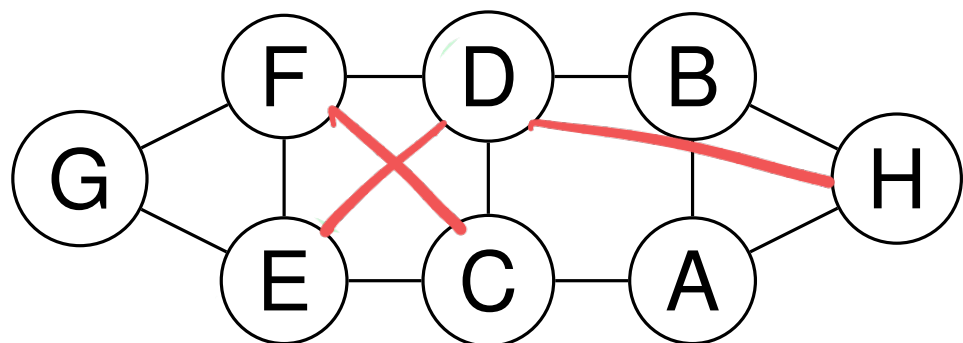
more than 2 vertices of odd degree

Why doesn't this have an Euler circuit?

Duplicate some of the edges so that every vertex is even degree. Using your additional edges, find an Euler circuit.

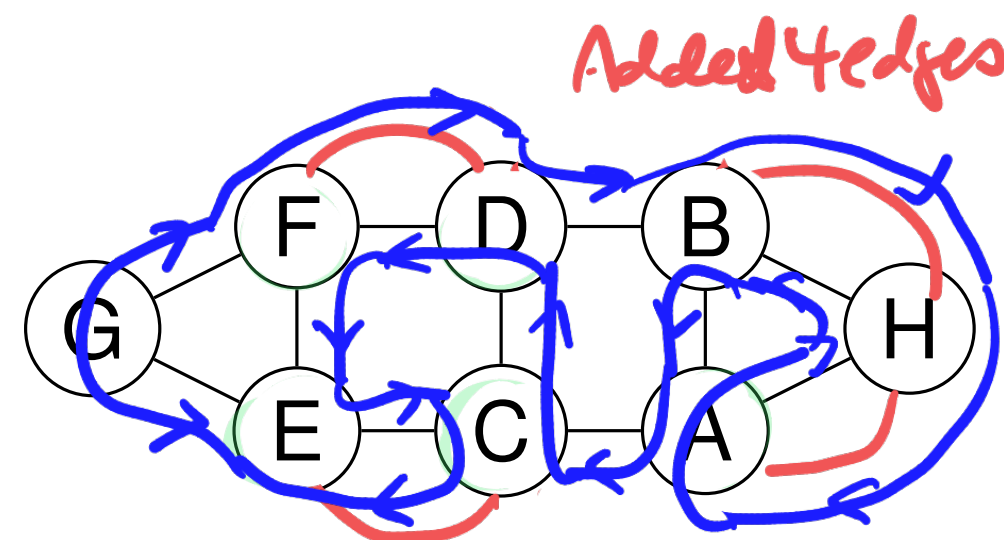


F G E F D C D B H A B A C E F

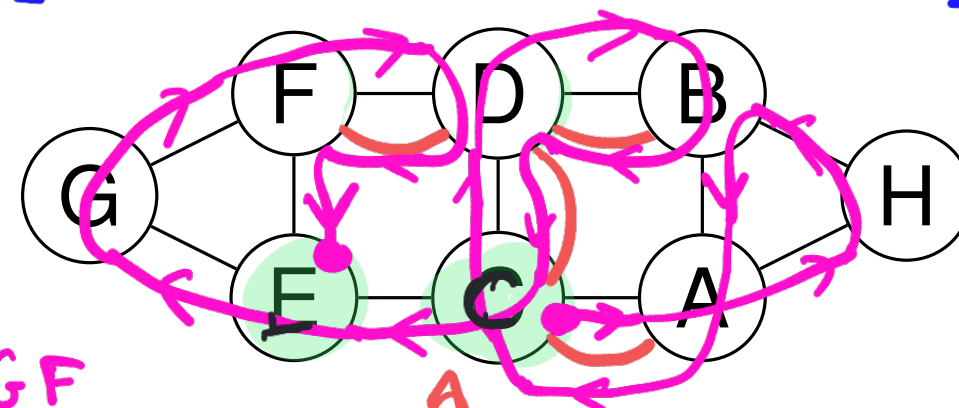


Not allowed! You can only duplicate edges that are already there!

C A H B A C
D B D C F G F
D F E

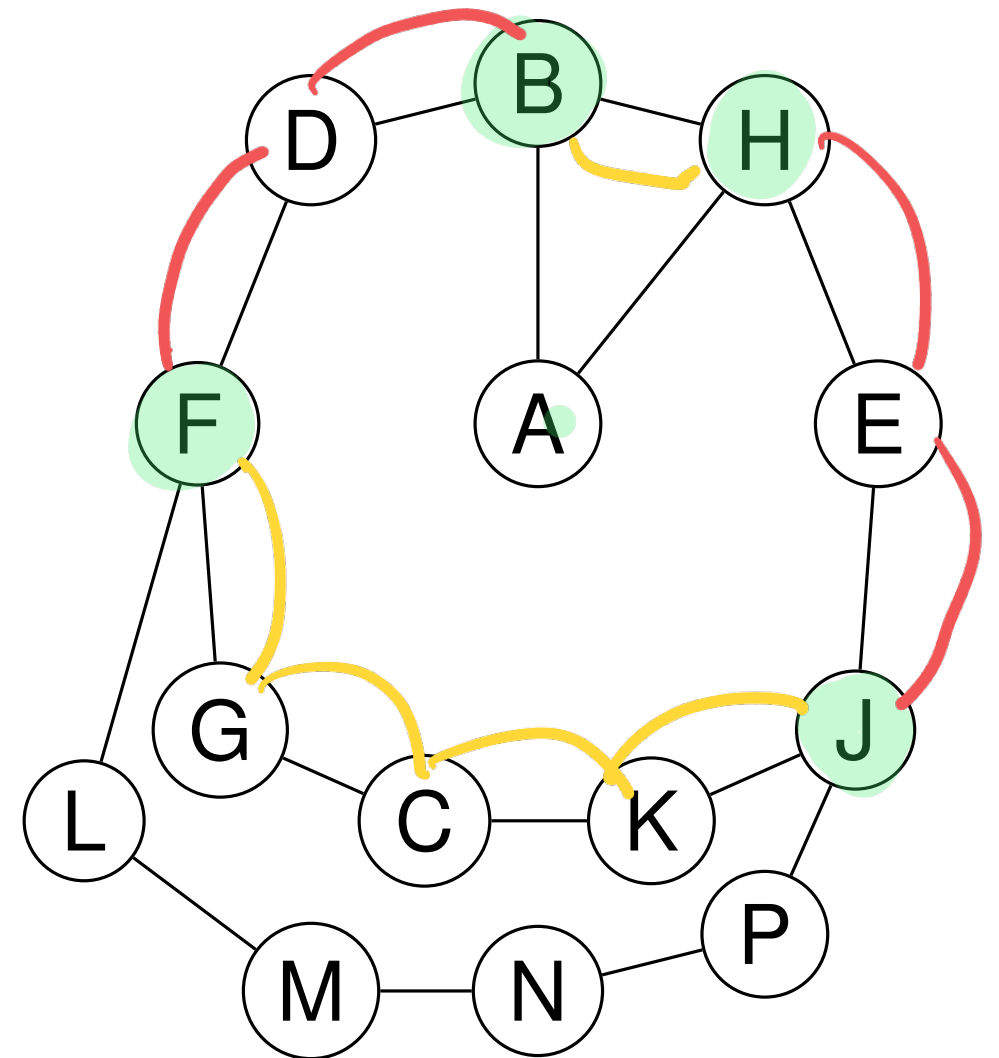
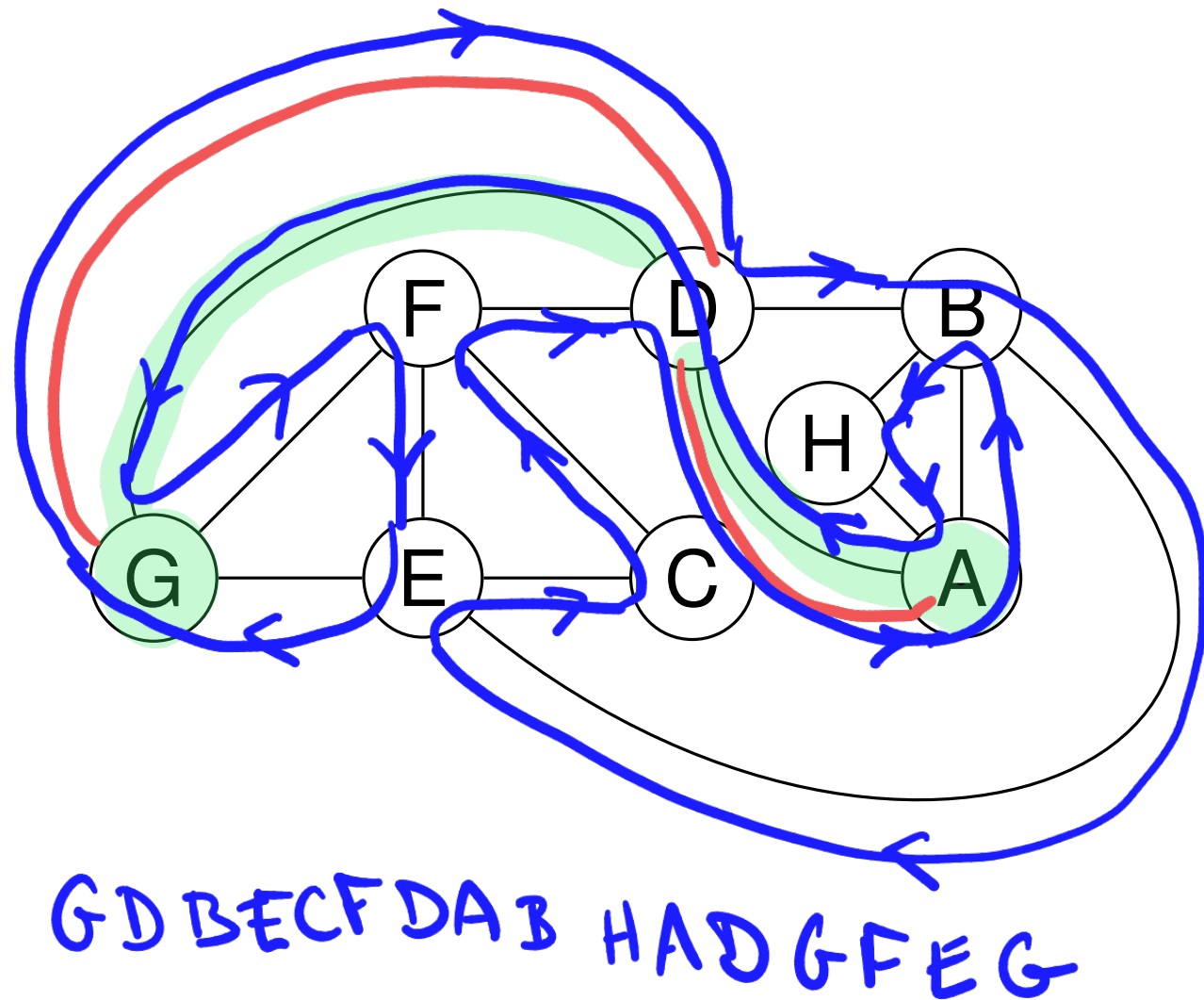


G F D B H A H B A C D F E C E G



odd degree ~~it~~
Euler path is possible here

Some other examples



Connect pairs of odd-degree
vtx with a path

Finding an optimal Eulerization can be tricky!

Definition:

eulerize a graph G :

Duplicate existing edges in a graph so that every vertex becomes even degree

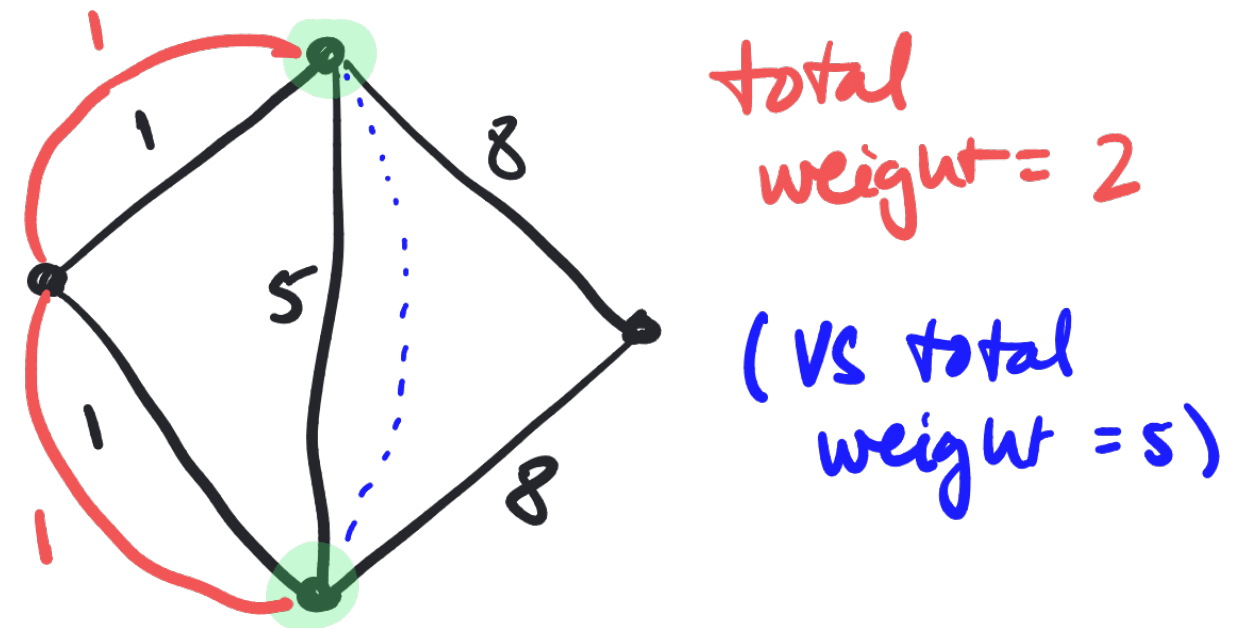
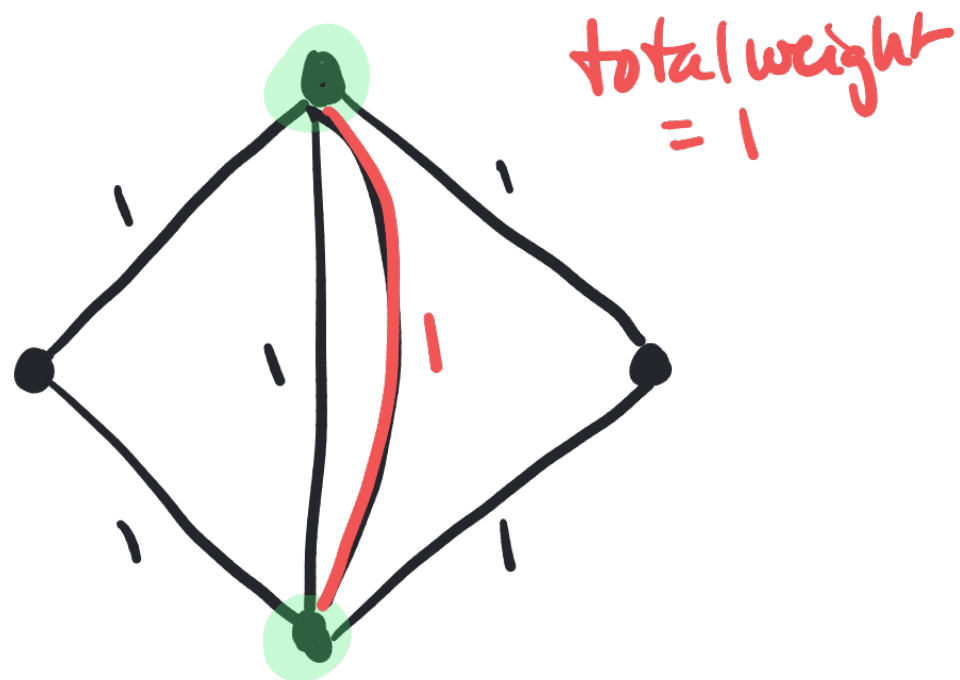
* always duplicate every edge

Optimal eulerization:

Eulerization with the fewest edges possible!

In a weighted graph, want sum of the added weights to be as small as possible

When is an optimal eulerization easy? Under what conditions do you think it is easy to obtain an optimal eulerization?



When is an optimal eulerization *easy*? Under what conditions do you think it is *easy* to obtain an optimal eulerization?

If there are exactly 2 odd-degree vtx, then you can duplicate a shortest (weighted) path, to get an optimal eulerization. Otherwise, not obvious what to do!